

Did you know?

As an early Open Source program, SPICE1 was widely distributed and used. It became so ubiquitous that "to SPICE a circuit" remains synonymous with circuit simulation. SPICE source code was from the beginning distributed by U.C. Berkeley for a nominal charge. The licence originally included distribution restrictions for countries not considered friendly to the USA, but the source code is currently covered by the BSD licence.

The real popularity of SPICE started with SPICE2 released in 1975. SPICE2, also coded in FORTRAN, was a much-improved program with more circuit elements, variable-timestep transient analysis using either the trapezoidal (second-order Adams-Moulton method) or the Gear integration method (also known as BDF), and an innovative FORTRAN-based memory allocation system.

A major improvement came about in March 1985 with the release of SPICE3 (also released under public domain). Written in 'C' language rather than FORTRAN, version 3 incorporated additional transistor types (metal semiconductor field-effect transistor, for example) and switch elements.

says, "The core simulator, SPICE, is augmented with high-quality device model characterisation, advanced layout extraction techniques, an integrated and highly productive user simulation environment, expert simulation CAD support, and interfaces to alternative simulators for use when their unique features are beneficial. The ability to run unlimited simulations without paying per-use licence fees supplements ADI's advantage, providing accurate and useful simulation results."

Advancements in IC technology, computer technology and simulation algorithms drive the evolution of ADICE and the IC simulation environment at ADI.

Choudhury adds, "ADICE, similar to SPICE simulations available in the open market, addresses the requirements of modern analogue/mixed-signal IC design, including fast, accurate simulation of huge circuits, layout extraction of devices and interconnects with regard to high-frequency electromagnetic effects, efficient handling of layout-dependent variation in small-geometry process nodes, and subtle modelling details needed for precision analogue circuits. The solution is a combination of advances in ADICE simulation engine, selective use of commercial simulators, careful modeling of transistors and other devices to meet the demands of high-performance analogue designers, expert use of layout extraction and electromagnetic modelling tools, and a holistic approach to

simulation application engineering."

HSPICE

HSPICE, Synopsys' internal proprietary simulator, is one of the most trusted and comprehensive circuit simulators in the market. It offers foundry-certified MOS device models with state-of-the-art simulation and analysis algorithms. With fully multi-threaded capabilities, HSPICE delivers additional performance using latest multicore computer hardware. CustomSim FastSPICE simulator delivers superior transistor-level verification performance and capacity for various classes of design, including custom digital, memory and analogue/mixed-signal circuits. It provides a comprehensive suite of analysis features that include circuit electrical-rule checking, electromigration, IR drop and MOS ageing analysis.

Hany Elhak, senior product marketing manager at Synopsys, says, "HSPICE is used extensively in standard cell libraries, mixed-signal ICs, and signal integrity analysis of high-speed

packages and boards. The leading performance of HSPICE is the result of 30 years of innovation in simulation and analysis algorithms optimised for multicore and distributed computing. As the industry moves to 16-nanometre processes, HSPICE is ready with the industry's fastest and most accurate FinFET support, offering BSIM-CMG models with simulation speed comparable to that of planar CMOS."

For full-chip mixed-signal verification, CustomSim is tightly coupled with Synopsys' VCS functional verification solution and CustomExplorer Ultra mixed-signal verification environment, providing a high-performance, advanced mixed-signal simulation, and regression debug and analysis solution.

PSpice

Cadence' PSpice is a full-featured, native analogue and mixed-signal circuit simulator. Used in conjunction with PSpice A/D, PSpice Advanced Analysis tools help designers improve yield and reliability of their designs. The term 'PSpice' is an acronym for 'personal simulation program with integrated circuit emphasis.'

Initially developed by MicroSim, PSpice is used in electronic design automation. MicroSim was bought by OrCAD, which was subsequently purchased by Cadence Design Systems.

PSpice has evolved into an analogue mixed-signal simulator. The software, now developed towards more complex industry requirements, is integrated in the complete system design flow in OrCAD and Cadence Allegro. It includes features such as circuit analysis with automatic opti-

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